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Mitosis, Meiosis, Cell Cycle Regulation, and Genetic Diversity

How two division pathways, and the checkpoints that police them, grow bodies, repair damage, and generate the variation that lets populations adapt.

01 Mitosis

Mitosis is the process of cell division that produces two genetically identical diploid daughter cells. It drives growth, tissue repair, replacement of damaged cells, and asexual reproduction in some organisms.

Prophase

- Chromosomes condense and become visible
- Nuclear envelope begins to break down
- Mitotic spindle starts forming

Metaphase

- Chromosomes align at the metaphase plate
- Spindle fibers attach to centromeres

Anaphase

- Sister chromatids separate
- Chromatids move toward opposite poles

Telophase

- Nuclear envelopes reform
- Chromosomes decondense

Cytokinesis

- Cytoplasm divides
- Two genetically identical diploid daughter cells are formed

WHY IT MATTERS

- Body growth
- Tissue repair
- Cell replacement
- Wound healing
- Maintains chromosome number (2n → 2n)

02 Meiosis

Meiosis is a specialized type of cell division that produces four genetically unique haploid gametes. It reduces chromosome number by half and creates genetic diversity required for sexual reproduction.

MEIOSIS I — HOMOLOGS SEPARATE

Prophase I

- Homologous chromosomes pair
- Crossing over occurs
- Genetic recombination begins

Metaphase I

- Homologous chromosome pairs align randomly

Anaphase I

- Homologous chromosomes separate

Telophase I

- Two haploid cells are produced

MEIOSIS II — SISTER CHROMATIDS SEPARATE

Prophase II

- Chromosomes condense again

Metaphase II

- Chromosomes align individually

Anaphase II

- Sister chromatids separate

Telophase II

- Four genetically different haploid cells are formed

SOURCES OF GENETIC VARIATION

- Crossing Over
- Independent Assortment
- Random Fertilization
- Mutation

WHY IT MATTERS

- Produces sperm and egg cells
- Maintains chromosome number between generations
- Increases genetic diversity
- Supports evolution and adaptation

03 Cell Cycle Regulation

The cell cycle runs G1 → S → G2 → Mitosis → Cytokinesis, with checkpoints acting as quality-control gates at each transition.

G1 CHECKPOINT

- Cell size
- Nutrient availability
- Growth factors
- DNA damage

S PHASE

- DNA replication occurs
- Sister chromatids are produced
- Centrosomes duplicate

G2 CHECKPOINT

- DNA replication completed
- DNA damage repaired
- Cell ready for mitosis

M CHECKPOINT (SPINDLE CHECKPOINT)

- Proper chromosome alignment
- Correct spindle attachment before chromosome separation

If checkpoints fail

- DNA mutations accumulate
- Chromosome segregation errors occur
- Uncontrolled cell division
- Tumor formation
- Cancer development

Mitosis vs. Meiosis

Feature	Mitosis	Meiosis
Number of Divisions	1	2
Daughter Cells	2	4
Chromosome Number	Maintained	Reduced by half
Genetic Makeup	Identical	Genetically different
Occurs In	Somatic cells	Germ cells
Purpose	Growth and repair	Sexual reproduction
Crossing Over	No	Yes

Key Facts

46	chromosomes (2n) in human somatic cells.	**23**	chromosomes (n) in human gametes.

S	phase is when DNA replication occurs.	**I**	Crossing over occurs during Prophase I.
I	Independent assortment occurs during Metaphase I.	•	Animal cells divide by a cleavage furrow.
□	Plant cells divide by forming a cell plate.		

04 Conclusion

Mitosis and meiosis are essential biological processes that work together to maintain life. Mitosis supports growth, tissue repair, and the replacement of damaged cells by producing genetically identical daughter cells. Meiosis produces genetically unique gametes through crossing over and independent assortment, creating the genetic diversity necessary for adaptation and evolution. This diversity improves the ability of populations to survive environmental changes and disease outbreaks.

> *In Pakistan, genetic diversity has helped researchers develop disease-resistant wheat and rice varieties that improve crop productivity, strengthen food security, and support sustainable agriculture.*

Together, mitosis and meiosis ensure the continuity of life while promoting healthy growth, successful reproduction, and long-term survival of species.

References

OpenStax. (2022). Biology 2e. OpenStax. https://openstax.org/details/books/biology-2e